🧩 What is an If-Else Statement?

The **if statement** is used to **test a condition**.
If the condition is **True**, a block of code runs.
If it's **False**, another block (in else) runs.

🧠 Basic Syntax

```
if condition:
    # code block (executes if condition is True)
else:
    # code block (executes if condition is False)
```

🔹 Example 1: Simple If-Else

```
age = 18

if age >= 18:
    print("You are eligible to vote.")
else:
    print("You are not eligible to vote.")
```

Output:

```
You are eligible to vote.
```

🔹 If without Else

You can use only if when you don't need an alternative path.

```
temperature = 35
if temperature > 30:
    print("It's a hot day.")
```

- **If-Elif-Else (Multiple Conditions)**

Use elif when you need to test more than one condition.

```
marks = 85


if marks >= 90:
    print("Grade: A+")
elif marks >= 75:
    print("Grade: A")
elif marks >= 50:
    print("Grade: B")
else:
    print("Grade: Fail")
```

Output:

Grade: A

- **Nested If (If inside If)**

Sometimes, you want to check a condition inside another condition.

```
age = 20
citizen = "Yes"


if age >= 18:
    if citizen == "Yes":
        print("Eligible to vote")
    else:
        print("Not a citizen")
```

```
else:
    print("Underage")
```

Output:

Eligible to vote

- **Short-hand (One-line) If**

If you have only one statement, you can write it in one line.

```
x = 10
if x > 5: print("x is greater than 5")
```

- **Short-hand If-Else (Ternary Operator)**

```
age = 17
print("Adult") if age >= 18 else print("Minor")
```

Output:

Minor

- **Logical Operators in If**

You can combine multiple conditions using:

- **and** → both must be True
- **or** → at least one True
- **not** → reverses the condition

```
age = 25
has_license = True

if age >= 18 and has_license:
    print("You can drive")
```

```
else:
    print("You cannot drive")
```

- **Comparison Operators in If**

Operator	Meaning	Example
==	Equal to	x == 10
!=	Not equal to	x != 5
>	Greater than	x > 5
<	Less than	x < 10
>=	Greater than or equal to	x >= 18
<=	Less than or equal to	x <= 20

- **Example: Check if a number is even or odd**

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

- **Example: Find the largest of three numbers**

```
a = 10
b = 25
c = 15
```

```
if a > b and a > c:
    print("A is largest")
elif b > a and b > c:
    print("B is largest")
else:
    print("C is largest")
```

🧠 **Summary**

Statement	Description
if	Executes if condition is True
elif	Checks next condition if previous is False
else	Executes when all above are False
and, or, not	Combine conditions
One-line if	if condition: statement
Ternary	x if condition else y